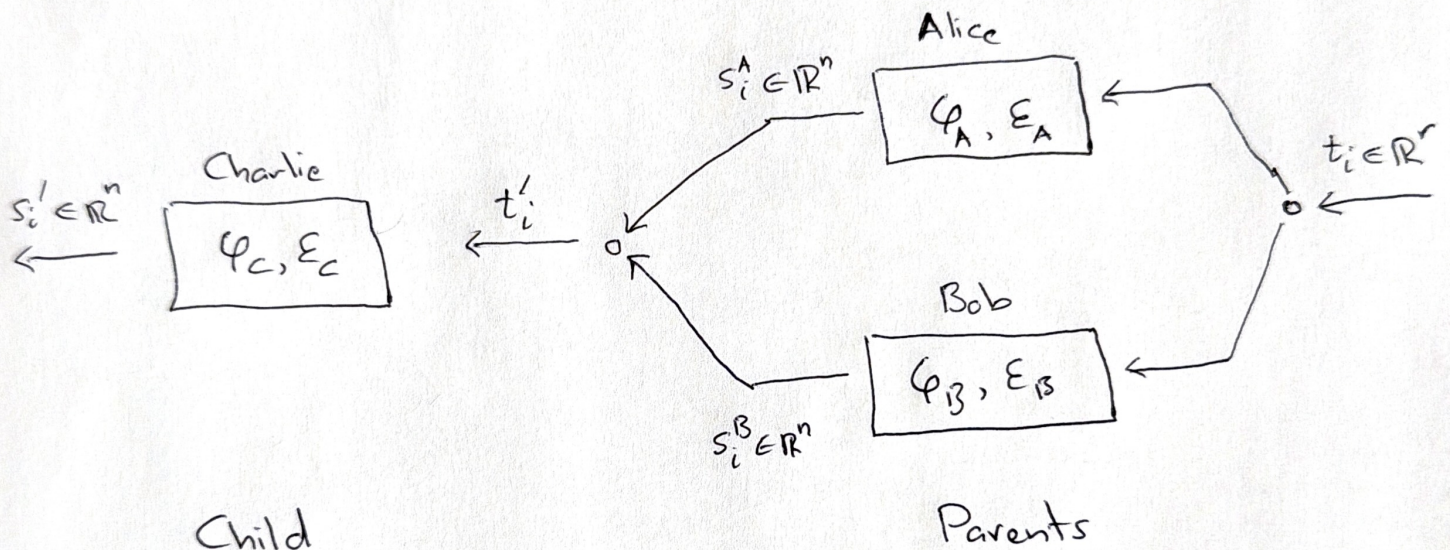


Further Noken genetic operators:

Let $u = (u_1, u_2, \dots, u_n)^T, v = (v_1, v_2, \dots, v_n)^T \in \mathbb{R}^n$

$$u \otimes v = (u_1 v_1, \dots, u_1 v_n, \dots, u_n v_1, \dots, u_n v_n)^T$$

$$u \circ v = (u_1 v_1, u_2 v_2, \dots, u_n v_n)^T$$



$$\varphi_C, E_C = \begin{cases} \varphi_A, E_A & \text{if } E_A < E_B \\ \varphi_B, E_B & \text{otherwise} \end{cases}$$

Kronecker

$$t_i^C \triangleq s_i^A \otimes s_i^B$$

Hadamard

$$t_i^C \triangleq s_i^A \circ s_i^B$$

Let $u = (u_0, \dots, u_{n-1})^T$, $v = (v_0, \dots, v_{n-1})^T \in \mathbb{R}^n$

For $k = 0, 1, \dots, n-1$

$$\text{define } (u \times v)_k = \sum_{i+j=k} u_i v_j$$

Convolution

$$t'_i \triangleq s_i^A \times s_i^B$$